

GANESHKUMAR V

☎ (+91) 78713 11800 | ✉ ganeshkv@iitgn.ac.in | 📷 Ganeshkumar-V | 🌐 ganeshkumar-v

EDUCATION

Indian Institute Technology - Gandhinagar

Gujarat, India

Ph.D. Mechanical Engineering (Thesis Submitted)

Aug. 2020 - present

CPI : 9.58/10 | Awarded Prime Minister's Research Fellowship (PMRF)

Anna University - Madras Institute of Technology (MIT)

Chennai, India

B.E Mechanical Engineering

Aug. 2016 - May 2020

CGPA : 9.84/10 | University Gold Medalist (Highest Academic Performance)

PUBLICATIONS

- 2025 **Journal of Propulsion and Power**
"Computational Study of Propulsive Performance of Frozen Nano-Aluminum and Water (ALICE) Mixtures", **Ganeshkumar V** and Dilip S Sundaram, [Link](#)
- 2025 **AIAA SciTech Forum 2025**
"Effects of Particle Size and Mixture Composition on Propulsive Performance of Gelled and Refrigerated Aluminum-Liquid Oxidizer Propellants", **Ganeshkumar V** and Dilip S Sundaram, [Link](#)
- 2025 **Combustion And Flame**
"A Computational Fluid Dynamics Model of Combustion of Nanoaluminum - Water Propellant Strands", Prasanna Kulkarni[†], **Ganeshkumar V**[†] and Dilip S Sundaram [[†]Equal Contribution] [Link](#)
- 2025 **Under Review - Journal of Propulsion and Power**
"Effect of Particle Agglomeration on Propulsive Performance of Aluminum-Water Propellant Rocket Motors", **Ganeshkumar V** and Dilip S Sundaram

PH.D. RESEARCH

A Computational Study on Propulsive Performance of Metal-Based Refrigerated and Gelled Composite Propellants

IIT Gandhinagar

Supervisor: Prof. Dilip Srinivas Sundaram

Dec. 2020 - Present

- ✓ **Propulsion Research:**
 - Developed a **high-fidelity CFD solver** using **OpenFOAM** to capture **multiphase flow dynamics** in **solid rocket motors**.
 - Optimized the solver and employed it to **quantify thrust, specific impulse**, and other performance parameters of metal-liquid oxidizer propellants.
 - Identified the **reasons for poor performance** observed in the experiments, quantified their effects, and proposed design guidelines for performance optimization
- ✓ **Multiphase Combustion Research:**
 - Developed and implemented a **multiphase combustion CFD framework** in OpenFOAM ("**regressionFoam**") to replicate **strand burner experiments**.
 - Predicted the **burning rates** of metal particles in liquid oxidizer mixtures and **validated** with the experiments.
 - Identified key factors influencing combustion performance and **formulated a hypothesis** to resolve discrepancies between experimental results and theoretical predictions.

ADDITIONAL RESEARCH PROJECTS

Computational Performance Evaluation and Optimization of Combustion Chemistry Calculations in OpenFOAM

IIT Gandhinagar

Supervisor: Prof. Dilip Srinivas Sundaram

Summer 2024

- Profiled **OpenFOAM's combustion libraries** and solver using **Intel VTune Profiler**, identifying major computational bottlenecks in gas-phase combustion simulations.
- Implemented **Hybrid MPI-OpenMP parallelization** strategies to accelerate chemical source term evaluations, achieving a **208x speedup** in computational performance.

Determination of Acoustic Resonance Modes in the Air Cavity

IIT Madras

Supervisor: Prof. K. Srinivasan

Summer 2019

- Conducted **Modal Acoustic Analysis** in ANSYS to determine the **resonance frequencies** of an air cavity resembling a combustion chamber of rocket engine and validated results against experimental data.

COMPUTING SKILLS

Programming & Parallel Computing: Object Oriented Programming, MPI, OpenMP, CUDA

Languages: C++, Python, MATLAB

Scientific Software & Tools: OpenFOAM, Cantera, Ansys-Fluent, Paraview

ACADEMIC ACHIEVEMENTS

- 2022 **Prime Minister's Research Fellowship (PMRF) - Government of India**, Awarded prestigious fellowship for conducting PhD research at IIT Gandhinagar
- 2020 **University Gold Medal - Anna University**, Secured highest CGPA among all the students in the Department of Mechanical Engineering
- 2019 **IIT Madras Summer Fellowship Programme**, Ranked among the top 20 students nationwide
- 20'17-20 **RNTBCI - AU Scholarship**, Awarded for academic excellence by Renault Nissan Technology & Business Centre India (RNTBCI) and Anna University (AU)

TEACHING EXPERIENCE

PMRF Teaching Fellow

Online

NPTEL - Ministry of Education, Government of India

Aug. 2022 - Present

- Conducted problem solving sessions for national level MOOC courses taught by IIT faculty.
- Supported hundreds of students nationwide in courses including Rocket Propulsion, Aeroengine Gas Turbine Cycles, CFD, Advanced Thermodynamics & Combustion, Heat Transfer and Numerical Methods.
- Facilitated deeper understanding on the topics through structured discussions; session recordings are available in the [YouTube](#) channel.

Graduate Teaching Fellow

Offline

Indian Institute of Technology - Gandhinagar

Aug. 2022 & Jan. 2024

- Independently tutored undergraduate and postgraduate students in Computational Fluid Dynamics and Fluid Dynamics courses.
- Designed and supervised innovative course projects and assignments in collaboration with instructors.
- Served as Lead Teaching Assistant for six courses, coordinating tutorials, assignments, and student engagement.